DESIGN MILESTONES

KAIRO
Outdoor Tech

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IDE 402

Milestone 1

RESEARCH

My initial research with AI focused on the history of water boilers, and I observed that their historical development has mainly been driven by functional improvements. For this reason, instead of selecting a specific beverage to focus on in my project, I aimed to identify the main issues of the core product through user research.

Afterward, I wanted to explore the contexts in which this product is used and noticed that products suitable for outdoor use are limited and not sufficiently developed in the market. During my research on existing products, I used AI to compare user reviews and product features, asking it to generate a benchmark analysis. This process helped me define my target area and design direction.

First (1891)

Faster boiler (1920)

Safer boiler (1950)

Existing products benchmark

During the analysis of benchmark, it was observed that there is a lack of products that both boil water quickly and offer easy setup. This gap was therefore identified as a key design target.

Interview Insights

Interview questions were prepared using AI and posed to campers. Based on the information gathered, the most frequently mentioned needs for the product concept were compiled, and a target user profile was created.

TARGET USERS

The primary users include outdoor enthusiasts, campers, and homeowners seeking **convenient hot water** solutions for outdoor activities and events.

USAGE SCENARIOS

Common scenarios involve camping trips, outdoor events, and home use, where users require **portability** and easy setup for varied conditions.

CHALLENGES

Key challenges include ensuring the heater is **weather-resistant**, lightweight for transport, and capable of efficient energy use in outdoor settings.

Sketches

Selected Design

Milestone 2

FEEDBACK-REJECTION

This step became the most critical in my process. After the feedback session on my latest design revealed that the concept was too superficial and did not result in a fully developed product. The user interaction was flawed, the materials were not suitable for outdoor use, the proportions between the user and the product were not as intended, and most importantly, an incomplete concept had led to an unfinished design.

I decided to reject the AI-generated steps at this stage and dive deeper into the topic to conduct thorough research, aiming for a fully coherent concept.

Milestone 3

RESEARCH

After the feedback session I received, I decided to further explore camping activities. While researching different types of camping and outdoor activities with the help of AI, one particular issue caught my attention: the environmental damage caused by the methods campers commonly use to boil water.

- camp fires
- gas stoves

This became a milestone for the project, and the direction of the design development changed significantly. I then deepened my research by focusing on the environmental impacts of these methods, and the research summary I developed is presented below.

Research Summary

How Campfires Affect Nature
Environmental Impact

Soil Damage

Campfires cause significant soil damage due to intense heat, which can lead to nutrient depletion and erosion, ultimately affecting the structure and health of the surrounding ecosystem.

Air Quality

Smoke from campfires contributes to air quality degradation, releasing harmful pollutants and particulates that can negatively impact both human health and the atmosphere, leading to broader environmental issues.

Wildlife Disruption

The presence of campfires can disrupt local wildlife, creating habitat loss and altering animal behaviors, which can result in imbalances within the ecosystem and affect biodiversity levels.

How gas stoves Affect Nature

40%

Emissions from gas stoves

Gas stoves used in trekking can produce 40% more carbon emissions compared to alternative methods, significantly impacting air quality and contributing to environmental degradation.

Key Needs for Alternatives

Reduce Carbon Footprint

Sustainable alternatives significantly lower greenhouse gas emissions, promoting a healthier environment while enabling trekkers to enjoy nature without contributing to climate change.

Minimize Fire Risk

Utilizing eco-friendly methods greatly reduces the risk of forest fires caused by traditional campfires and harmful emissions from gas stoves.

Preserve Natural Landscapes

Sustainable practices protect biodiversity and maintain pristine environments, ensuring that future generations experience the beauty of untouched nature during their outdoor adventures.

Milestone 4

PROTOTYPE

After conducting research and developing a research summary, I aimed to design a portable, lightweight, and durable water heating system that enables fast boiling through induction heating technology, can be used without harming nature, follows the design language of outdoor equipment, and features a simple and intuitive interface.

Through iterative generating processes and prompt refinements, I finalized this design direction. Below, I visualized the key design decisions that I specifically focused on.

PROMPT(GEMINI)

portable trekking kettle-thermos hybrid, 500ml capacity, compact cylindrical form, lightweight outdoor gear design, durable stainless steel body with insulated outer layer, minimal transparent glass window to check water level, designed for hikers and backpackers, removable lid that seals tightly to keep water at boiling temperature inside a backpack, modern rugged industrial design, matte dark finish with subtle outdoor accents, ergonomic proportions, minimal interface, designed to boil water when placed on an induction heating base, realistic industrial design concept sketch, clean product visualization, high detail portable battery powered induction heating base for outdoor use, compact circular platform, rugged minimal design, matte anodized aluminum body, small battery indicator lights, lightweight trekking equipment aesthetic, designed to heat a kettle placed on top, durable and weather resistant, modern industrial design render

Design Decisions

Branding

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Milestone 5

PROTOTYPE

During the AI prototyping process, it was decided to transform this design into a complementary product family. The goal of the system was to generate energy through sustainable sources without harming nature and to power a water-boiling thermos that required supporting products to complete the ecosystem.

The design focused on utilizing available energy sources such as solar power, motion-based energy (rivers and wind), and kinetic energy generated by campers themselves. In addition to boiling water, these energy sources were also intended to support cooking and general heating, leading to the creation of a fully integrated outdoor experience.

PROMPT(GEMINI)

Create a cohesive outdoor product family based on an existing sustainable trekking water boiler design.

The product family must share the SAME design language, proportions, materials, and visual identity as the main product.

Main product reference:
A portable outdoor induction water boiler designed for trekking hikers. The system uses collectible renewable energy (solar and kinetic energy) stored in a battery-powered induction base to boil water or heat food. The design is minimalist, lightweight, durable, and sustainable.

Design Language Requirements:
- Minimalist industrial design
- Compact cylindrical geometry
- Soft rounded edges
- Lightweight trekking equipment aesthetic
- Rugged but clean appearance
- Modular ecosystem logic
- Matte finishes
- Outdoor premium look
- Functional and intuitive forms

Materials (CMF consistency across all products):
- Stainless steel inner structures
- Insulated outer shells
- Matte anodized aluminum or recycled polymer exterior
- Small controlled transparent window details (minimal glass usage)
- Neutral colors: matte black, dark gray, sand, with subtle outdoor accent color (muted orange or earth tone)

Create a COMPLETE PRODUCT FAMILY including:

1. Water Storage & Purification Container
- Portable water collection container
- Integrated filtration system
- Designed to collect water from rivers or lakes
- Compatible visually and proportionally with the boiler system

2. Food Heating Container
- Modular food container
- Stackable design
- Lightweight trekking cooking solution

3. Deployable Solar Energy System
- Foldable or rollable solar panels
- Portable trekking scale
- Designed to attach to backpack or camp setup
- Charges the energy storage system

4. Kinetic Energy Generator
- Kinetic energy with turning handle
- Stores energy while hiking

5. Power Bank Module
- Energy storage unit using collected renewable energy
- Powers devices and the induction system
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General Requirements:
- All products must look like part of one unified ecosystem
- Maintain proportional harmony and shared visual DNA
- Show modular compatibility between products
- Outdoor trekking usage context
- Industrial design concept visualization
- High detail, realistic product renders
- Clean studio or minimal outdoor presentation

Milestone 6

TESTING

AI simulations were conducted to evaluate these sustainable energy generation and storage designs within a camping environment, including calculations of their daily energy outputs.

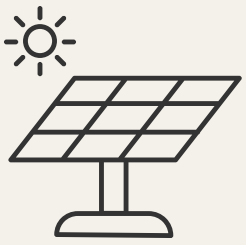
The results confirmed that the products function as complementary systems, enabling continuous energy generation and storage over a 24-hour period.
Energy contributions of the systems (approx.):

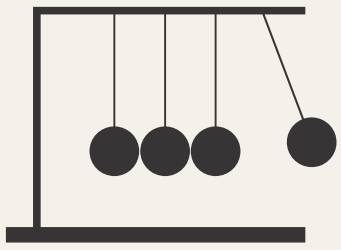
- Solar system: ~500 Wh per hour in sunny conditions (primary daytime energy source)
- Hydro turbine: ~290 Wh per hour continuous generation (day and night base energy)
- Hand-crank kinetic generator: ~42 Wh per hour during active use (emergency and support energy)

Collectible Energy

ENERGY EFFICENCY

The designed service, will generate energy in nature in a sustainable way. In this way, energy will be provided for outdoor activities without causing harm to the environment. Such as kinetic energy, battery power, and solar energy.





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Collecting Energy

Portable Solar panels
Collecting solar energy on sunny days

Kinetic Energy Handle
For urgent situations

Kairo aerohyrdo turbine
Wind turbine system for energy collection while trekking.

Kairo aerohyrdo turbine
Mounted at height in camping areas to generate and store wind energy.

Kairo aerohyrdo turbine
If there is a water flow near the campsite, it can be fixed with a tripod, generating energy as the water current spins the turbine.

Milestone 7

COMMUNICATE

Final research was conducted to improve the communication between the entire designed service and campers. During the camp simulation, several gaps were identified, particularly the lack of interaction between the products and the users. Since multiple devices operate simultaneously, difficulties were observed in tracking the energy collected by the systems and monitoring the charge levels of the induction stoves.

To address this issue, a web application was designed to enable easy monitoring of the energy generated by all technological devices within the system and the charge status of the induction stoves. With its simple interface, the web application strengthens the interaction between campers and the products. Additionally, several illustrated usage scenarios were created to clearly communicate how the products are intended to be used.

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Mobile App